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31 March 1985
SUPERSEDING
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10 November 1969

MILITARY SPECIFICATION

TEST SET, SUBASSEMBLY MX-8050/TPM-22A
(Printed Circuit Card Tester)

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Test Set, Subassembly MX-8050/TPM-22A hereinafter referred to as printed circuit card tester, part of Radar Test Set AN/TPM-22A. The printed circuit card tester is used to verify proper operation of printed circuit cards in Radar Set AN/VPS-2 and Radar Test Set AN/TPM-22A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of

FSC 4931

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. AMSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-46357	-	Computer, Gun Direction: M18 Manufacturing Processes and Inspection of
MIL-C-48436	-	Test Set Subassembly: 11726378 (7.5 MHz Clock)
MIL-R-50318	-	Regulator Voltage: 10548034/11727766
MIL-R-50319	-	Regulator Voltage: 10548043/11727763
MIL-T-50336	-	Test Set Subassembly: 10554898
MIL-T-50337	-	Test Set Subassembly: 10558483
MIL-T-50338	-	Test Set Subassembly: 11726303
MIL-T-50340	-	Test Set Subassembly: 11726333
MIL-T-50341	-	Test Set Subassembly: 11726335
MIL-T-50342	-	Test Set Subassembly: 11726337
MIL-T-50343	-	Test Set Subassembly: 11726338
MIL-T-50344	-	Counter, Electronic: 10548009

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other government documents, drawings, and publications. The following other Government drawings form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

F10548009	-	PRF Counter

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F10548034	-	± 15V Regulator
F10548043	-	± 25V Regulator
F10554898	-	Pulse Generator
B10554918	-	Adapter Assembly (CP37)
C10556106	-	Connector, Adapter, TPS
F10558483	-	Card Reader Assembly
F11726303	-	Test Socket Assembly
F11726306	-	Test Set, Subassembly MX-8050/TPM-22A
F11726333	-	Line Driver A
F11726335	-	Line Driver B/C
F11726337	-	Line Driver D
F11726338	-	Low Voltage Filter
D11726378	-	7.5 MHz Clock
D11726394	-	Adapter 1CP4
D11726396	-	Adapter 1CP6
B11726471	-	Test Cards for Test Set Subassembly MX-8050/TPM-22A
C11727743	-	Cable Assembly, W85
F11727763	-	± 25V Regulator
F11727766	-	± 15V Regulator
D11727789	-	Power Supply, Regulated, DC
B11740292	-	Adaptor Assembly (CP32)
F11829942	-	Generator, Signal

PACKAGING DATA SHEET

11726306	-	Packaging of Test Set, Subassembly MX-8050/TPM-22A

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

REQUIREMENTS

3.1 Fabrication. The printed circuit card tester shall be manufactured in accordance with Drawing F11726306 and drawings pertaining thereto. The subassemblies of the printed circuit card tester shall have been inspected and accepted in accordance with the drawing prior to assembly in the printed circuit card tester. The following subassemblies are covered by the indicated specifications:

Unit No.	Name	Specification
1A3A2A1A1;	Test Set Subassembly (Card Reader Assy):	10558483; MIL-T-50337
1A3A2A1A2;	Test Set Subassembly (Test Socket Assy):	11726303; MIL-T-50338
1A3A2A1A3;	Generator, Signal:	11829942
1A3A2A1A5;	Test Set Subassembly (Line Driver A):	11726333; MIL-T-50340
1A3A2A1A6;	Test Set Subassembly (Line Driver B/C):	11726335; MIL-T-50341
1A3A2A1A7;	Test Set Subassembly (Line Driver B/C):	11726335; MIL-T-50341
1A3A2A1A8;	Test Set Subassembly (Line Driver D):	11726337; MIL-T-50342
1A3A2A1A9;	Test Set Subassembly (Low Voltage Filer):	11726338; MIL-T-50343
1A3A2A1A10;	Counter, Electronic:	10548009; MIL-C-50344
1A3A2A1A11;	Test Set Subassembly (Pulse Generator):	10554898; MIL-T-50336
1A3A2A1A12;	Regulator, Voltage:	10548034/11727766; MIL-R-50318
1A3A2A1A13;	Regulator, Voltage:	10548043/11727763; MIL-R-50319
1A3A2A1A1Z1;	Test Set Subassembly (Clock, 7.5MHz):	11726378; MIL-C-48436

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3.2 General specifications. Where applicable, the printed circuit card tester shall meet the requirements of MIL-F-13926 and MIL-C-46357 as follows:

3.2.1 MIL-F-13926

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishings

d. Part identification and marking

e. Workmanship

f. Soldering electrical connections

3.2.2 MIL-C-46357

a. Etched circuit assemblies

b. Wiring

c. Threaded fasteners

d. Marking

e. Finishes

3.3 Environmental service conditions.

3.3.1 Vibration. The printed circuit card tester shall be mounted on the vibration table in normal service position. The printed circuit card tester shall meet the requirements of 3.4 and 3.5 after being subjected to continuous vibration for two hours along each of three mutually perpendicular axes within the frequency range and amplitude as follows:

Frequency Range (Hz ± 2 Percent)	Excursion (double amplitude) (inches ± 10 percent)
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 55	0.02 inch

The rate of change of frequency at each frequency range shall not exceed one cycle per second per second (1 cps/sec). Vibration tests shall be conducted under ambient room conditions.

3.3.2 Shock. The printed circuit card tester shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after being subjected to 18 impact shocks, consisting of three shocks in opposite directions along each of the three mutually perpendicular axes. The shock impulses shall have an acceleration

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of 15 gravity units (G's) and a pulse duration of 11 ± 1 milliseconds.

3.3.3. Altitude.

3.3.3.1 Operating. The printed circuit card tester shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.3.2 Non-operating. The printed circuit card tester shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The printed circuit card tester shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between +32°F and +131°F.

3.3.4.2 Non-operating (storage). The printed circuit card tester shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 48 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The printed circuit card tester shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance.

3.5.1 POWER/ON/OFF switch S1.

3.5.1.1 POWER/OFF legend. The POWER/OFF legend of S1 shall be illuminated when 24 Vdc ±5% is applied to J1-U(+) and J1-T(Ret) and S1 is in the normally closed (OUT) position.

3.5.1.2 POWER/ON legend. The POWER/ON legend of S1 shall be illuminated when 24 Vdc ± 5% is applied to J1-U(+) and J1-T(Ret) and S1 is in the normally open (IN) position.

3.5.2 Circuit breakers. Circuit breakers CB1 through CB8 shall be closed when their respective reset buttons are depressed so white bands are not exposed. Each circuit breaker shall be open when its reset button is pulled out manually to expose white band, and shall open automatically within 20 seconds when the indicated current overload is passed through it:

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Circuit Breaker	Overload Current
-25V (CB1)	2 Adc
+25V (CB2)	2 Adc
-15V (CB3)	2 Adc
+15V (CB4)	2 Adc
+6.6V (CB5)	6 Adc
+24V EXT (CB6)	2 Adc
+24V INT (CB7)	6 Adc
LINE (CB8)	2 Aac

3.5.3 LEVEL/AC/DC switch S2.

3.5.3.1 LEVEL/DC legend. The LEVEL/DC legend of S2 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.1.2 are met.

b. 24 Vdc ± 5% applied to J1-S (+) and J1-T (Ret).

c. +24V INT circuit breaker (CB7) closed (white band not visible).

d. S2 in normally closed (OUT) position.

3.5.3.2 LEVEL/AC legend. The LEVEL/AC legend of S2 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S2 in normally open (IN) position.

3.5.4 OVERLOAD/OFF/ON switch S7.

3.5.4.1 OVERLOAD/OFF legend. The OVERLOAD/OFF legend of S7 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S7 in normally closed (OUT) position.

3.5.4.2 OVERLOAD/ON legend. The OVERLOAD/ON legend of S7 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S7 in normally open (IN) position.

3.5.5 PROFILE/OFF/ON switch S3.

3.5.5.1 PROFILE/OFF legend. The PROFILE/OFF legend of S3 shall be illuminated when all of the following conditions are met:

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a. Conditions of 3.5.3.1a through 3.5.3.1c. are met.

b. S3 in normally closed (OUT) position.

3.5.5.2 PROFILE/ON legend. The PROFILE/ON legend of S3 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S3 in normally open (IN) position.

3.5.6 NORMAL/HIGH/LOW switch S4.

3.5.6.1 NORMAL legend. The NORMAL legend of S4 shall be illuminated when all of the conditions stated in 3.5.5.1 are met.

3.5.6.2 HIGH legend. The HIGH legend of S4 shall be illuminated when all of the following conditions are met:

a. PROFILE/ON legend of S3 illuminated in accordance with 3.5.5.2.

b. S4 in normally closed (OUT) position.

3.5.6.3 LOW legend. The LOW legend of S4 shall be illuminated when all of the following conditions are met:

a. PROFILE/ON legend of S3 illuminated in accordance with 3.5.5.2.

b. S4 in normally open (IN) position.

3.5.7 TEST/1/2 switch S5.

3.5.7.1 TEST/1 legend. The TEST/1 legend of S5 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S5 in normally closed (OUT) position.

3.5.7.2 TEST/2 legend. The TEST/2 legend of S5 shall be illuminated when all of the following conditions are met:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. are met.

b. S5 in normally open (IN) position.

3.5.8 RESET/OFF/ON switch S6.

3.5.8.1 RESET/OFF legend. The RESET/OFF legend of S6 shall be illuminated when all of the following conditions are met:

a. Conditions 3.5.3.1a. through 3.5.3.1c. are met.

b. Self Test Card ST 1 properly inserted in card reader assembly and

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handle of card reader assembly thrown to the right-hand position. Adapter CP6 inserted in test sockets A and B.

c. S6 in normally closed (OUT) position.

3.5.8.2. RESET/ON legend. The RESET/ON legend of S6 shall be illuminated under the following conditions:

a. Conditions of 3.5.3.1a. through 3.5.3.1c. and 3.5.8.1b. are met.

b. S6 in normally open (IN) position.

3.5.9 Power indicators. The power indicator lamps, +6.6V, -15V, +15V, +24V EXT, +24V INT, -25V, +25V and LINE shall be illuminated under the following conditions:

a. Conditions of 3.5.3.1a, 3.5.3.1b., and 3.5.8.1b. are met.

b. Circuit breakers CB1 through CB8 closed.

c. 33 Vdc ±1% applied between J1-A(-) and J1-B(Ret).

d. 33 Vdc ±1% applied between J1-C(+) and J1-D(Ret).

e. 23 Vdc ±1% applied between J1-E(-) and J1-F(Ret).

f. 23 Vdc ±1% applied between J1-G(+) and J1-F(Ret).

g. 5.9 Vdc ±1% applied between J1-P(+) and J1-R(Ret).

h. 115 Vac ±10%, 400 ±5 Hz applied between J1-V and J1-W.

3.5.10 Digiswitch lamps. Two TEST POINT A and two TEST POINT B thumbwheel selector lamps shall be illuminated under conditions stated in 3.5.8.1a. and 3.5.8.1b. Reading from left to right, the first switch in each group shall be illuminated and will indicate which other switch in the group is active (illuminated) as follows:

No. illuminated in first switch	Other switch illuminated
0	second
1	third
2	fourth
3	fifth
4	sixth
5	seventh

3.5.11 Test points A and B, and TEST IDENTIFICATION lamp outputs. The outputs listed in Table I shall appear at test points A and B, and the TEST IDENTIFICATION lamps shall be illuminated as given in Table I, under the following conditions:

a. Conditions of 3.5.3.1a., 3.5.3.1b., and 3.5.9b. through 3.5.9h. are met (with exceptions noted in Table I.).

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b. Front panel switches are set as follows (with exceptions noted in Table I):

Switch	Setting
LEVEL/AC/DC	LEVEL/DC
OVERLOAD/OFF/ON	OVERLOAD/OFF
PROFILE/OFF/ON	PROFILE/OFF
NORMAL/HIGH/LOW	NORMAL
TEST/1/2	TEST/1
RESET/OFF/ON	RESET/OFF

c. Test cards listed in Table I are properly inserted in the card reader and the handle is thrown to the right-hand position.

d. Adapter CP6 inserted in test sockets A and B for test cards ST1 through ST33, and adapter CP4 inserted in test sockets B and C for test cards ST34 and ST 35.

The voltage levels of pulses shall be $+3.5\ ^{+2.0}_{-1.5}$ for a logic 1 and $0.0\ ^{+0.4}_{-0.8}$ for logic 0. Pulse outputs shown in Figure 1 are signals at TEST POINT A in relation to signal at TEST POINT B.

Where more than one pin number is given in Table I under Test Point Selector A or B, each pin so numbered shall give the same result. For example, with the first test given under Test Card ST7, any combination of 02, 04, 06, or 08 on Test Point Selector A, and 02 on Test Point Selector B shall yield the same output.

3.6 First Article. Unless otherwise specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.7 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Unless otherwise specified in the item detail specification, or contract, the following types of inspection shall be conducted on the unit or product:

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a. First article inspection (4.3)

b. Quality conformance inspection (4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article as designated by the contracting officer (see 6.2) for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of 1 test set.

4.3.2 Inspection to be performed. First article sample may be subjected by the Government to any or all of the examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component, or test specimen in the sample to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II and III and methods of inspecition in 4.7. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein. Workmanship shall be in accordance with MIL-STD-454 and other applicable specifications.

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TABLE I. Self test card output.

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 1	ST,A,1	DVM	01	-	Test/1/2 : Test/2	-0.4 to +0.4 Vdc	1,8
ST 1	ST,A,1	DVM	01-56	-	-	+3.8 to +5.2 Vdc	1,8
ST 1	ST,A,1	DVM	-	01-56	-	+3.8 to +5.2 Vdc	3,16
ST 2	ST,C,2	DVM	01-56	-	Test/1/2 : Test/2	+3.8 to +5.2 Vdc	1,8
ST 3	ST,1,2, 128	O-Scope	01-08	01	-	T pulse (fig. 1)	8,13
ST 4	ST,4,64	O-Scope	01-08	01	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 5	ST,1,4	O-Scope	01-08	01	-	T pulse (fig. 1)	8,13
ST 6	ST,2,4	O-Scope	01-08	01	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 7	ST,1,2, 4	O-Scope	02,04 06,08	02	-	T pulse (fig. 1)	8,13
ST 7	ST,1,2 4	O-Scope	12	12	-	$\overline{\text{STROBE}}$ pulse (fig. 1)	8,13
ST 7	ST,1,2, 4	O-Scope	19	19	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 7	ST,1,2, 4	O-Scope	41	41	-	STROBE pulse (fig. 1)	8,13
ST 7	ST,1,2, 4	O-Scope	43	43	-	7.5 MHz clock (fig. 1)	8,12, 13
ST 8	ST,8	O-Scope	02,04 06,08	02	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	19	19	-	T pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	37	47	-	F1 pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	39	47	-	$\overline{F1}$ pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	41	47	-	F2 pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	43	47	-	$\overline{F2}$ pulse (fig. 1)	8,13
ST 8	ST,8	O-Scope	47	47	-	$\overline{F4}$ pulse (fig. 1)	8,13
ST 9	ST,1,8	O-Scope	02,04 06,08	02	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 10	ST,2,8	O-Scope	02,04 06,08	02	-	$\overline{T}$ pulse (fig. 1)	8,13

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 11	ST,1,2 8	O-Scope	01,02 03,04	01	-	$\overline{T}$ pulse (fig. 1)	8,13
ST 11	ST,1,2 8	O-Scope	05	05	LEVEL/AC/DC : LEVEL/AC FREQ : X1-KHz FREQ dial : 1	0 to 3.0 V pk-pk min 1 KHz (adjust with LEVEL control)	8,13, 14
ST 11	ST,1,2, 8	DVM	05,06 07,08	24	-	-1.4 to +1.4 Vdc (adjust LEVEL control)	2,9
ST 11	ST,1,2, 8	DVM	30	24	-	+4.5 to +5.5 Vdc	2,9
ST 11	ST,1,2, 8	DVM	31	25	PROFILE/ON/OFF : PROFILE/ON NORMAL/HIGH/LOW : HIGH	+5.0 to +6.0 Vdc	2,9
ST 11	ST,1,2 8	DVM	32	26	PROFILE/ON/OFF : PROFILE/ON NORMAL/HIGH/LOW : LOW	+4.0 to +5.0 Vdc	2,9
ST 12	ST,4,8	DVM	02	01	-	-9.0 to -11.0 Vdc	3,9
ST 12	ST,4,8	DVM	03	01	-	+2.5 to +3.5 Vdc	2,9
ST 12	ST,4,8	DVM	04	01	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	04	05	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	04	06	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	04	07	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	04	08	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	04	28	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	25	28	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	14	28	-	-14.85 to -15.15 Vdc	2,9
ST 12	ST,4,8	DVM	24	28	-	+21.0 to +27.0 Vdc	2,9
ST 12	ST,4,8	DVM	36	34	-	-14.85 to -15.15 Vdc	2,9

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 12	ST,4,8	DVM	37	35	-	+14.85 to +15.15 Vdc	2,9
ST 12	ST,4,8	DVM	20	22	-	-14.85 to -15.15 Vdc	2,9
ST 12	ST,4,8	DVM	21	23	-	-14.85 to -15.15 Vdc	2,9
ST 12	ST,4,8	DVM	49	51	-	-24.75 to -25.25 Vdc	2,9
ST 12	ST,4,8	DVM	50	52	-	-24.75 to -25.25 Vdc	2,9
ST 12	ST,4,8	DVM	55	53	-	+24.75 to +25.25 Vdc	2,9
ST 13	ST,1,4,8	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1	1 KHz sine waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 13	ST,1,4,8	O-Scope	58	58	FREQ : X1-KHz FREQ dial : 1	1 KHz sine waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 14	ST,2,4,8	DVM	55	01	MATRIX SELECTOR 1 : 12	+14.85 to +15.15 Vdc	2,9
ST 14	ST,2,4,8	DVM	56	01	MATRIX SELECTOR 2 : 12	+14.85 to +15.15 Vdc	2,9
ST 14	ST,2,4,8	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1	1 KHz triangle waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 14	ST,2,4,8	O-Scope	58	58	FREQ : X1-KHz FREQ dial : 1	1 KHz triangle waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 15	ST,1,2,4,8	DVM	55	01	MATRIX SELECTOR 1 : 12	-14.85 to -15.15 Vdc	2,9
ST 15	ST,1,2,4,8	DVM	56	01	MATRIX SELECTOR 2 : 12	-14.85 to -15.15 Vdc	2,9
ST 15	ST,1,2,4,8	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1	1 KHz square waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 15	ST,1,2,4,8	O-Scope	58	58	FREQ : X1-KHz FREQ dial : 1	1 KHz square waveform, 1.4 to 2.0 V pk-pk	8,13,14
ST 16	ST,16	DVM	55	01	MATRIX SELECTOR 1 : 12	+21.0 to +27.0 Vdc	2,9

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 16	ST,16	DVM	56	01	MATRIX SELECTOR 2 : 12	+21.0 to +27.0 Vdc	2,9
ST 16	ST,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1	1 KHz negative ramp waveform 0.60 to 1.1 V	8,13, 14
ST 16	ST,16	O-Scope	58	58	FREQ : X1-KHz FREQ dial : 1	1 KHz negative ramp waveform 0.60 to 1.1 V	8,13, 14
ST 17	ST,1,16	DVM	55	01	MATRIX SELECTOR 1 : 12	Variable, 0 to +36 Vdc	2,5,9, 10,15
ST 17	ST,1,16	DVM	56	01	MATRIX SELECTOR 2 : 12	Variable, 0 to +36 Vdc	2,5,9, 10,15
ST 17	ST,1,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION: ∿	1 KHz sine waveform 0 to 30 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14
ST 17	ST,1,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION: ⊓	1 KHz square wave form 0 to 30 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14
ST 17	ST,1,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION : △	1 KHz triangle wave-form, 0 to 30 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14
ST 17	ST,1,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION : ⊿	1 KHz positive ramp waveform, 0 to 15 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14
ST 17	ST,1,16	O-Scope	57	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION : ⊾	1 KHz negative ramp waveform, 0 to 15 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 17	ST,1,16	O-Scope	58	57	FREQ : X1-KHz FREQ dial : 1 FUNCTION : ~	1 KHz sine waveform 0 to 30 V pk-pk min. adjustable by AMPLITUDE control	8,13, 14
ST 18	ST,2,16	DVM	01-07	-	-	+3.8 to +5.2 Vdc	1,8
ST 18	ST,1,16	DVM	01	-	RESET/OFF/ON : RESET/ON	-0.4 to +0.4 Vdc	1,8
ST 18	ST,1,16	DVM	08	-	MATRIX SELECTOR 2 : 12	+7.7 to +9.7 Vdc	1,8
ST 19	ST,1,2, 16	DVM	57	01	-	45 to 55 ohms	2
ST 19	ST,1,2, 16	DVM	58	01	-	45 to 55 ohms	2
ST 19	ST,1,2, 16	DVM	55	01	MATRIX SELECTOR 1 :		2
					1	Infinite ohms	
					2	90 to 110K ohms	
					3	67.5 to 82.5K ohms	
					4	29.7 to 36.3K ohms	
					5	9 to 11K ohms	
					6	6.75 to 8.25K ohms	
					7	2.97 to 3.63K ohms	
					8	900 to 1100 ohms	
					9	675 to 825 ohms	
					10	297 to 363 ohms	
					11	90 to 110 ohms	
					12	Less than 1 ohm	

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 19	ST,1,2, 16	DVM	56	01	MATRIX SELECTOR 2 :		2
					1	Infinite ohms	
					2	90 to 110K ohms	
					3	67.5 to 82.5K ohms	
					4	29.7 to 36.3K ohms	
					5	9 to 11K ohms	
					6	6.75 to 8.25K ohms	
					7	2.97 to 3.63K ohms	
					8	900 to 1100 ohms	
					9	675 to 825 ohms	
					10	297 to 363 ohms	
					11	90 to 110 ohms	
					12	Less than 1 ohm	
ST 19	ST,1,2, 16	DVM	01	02	-	Less than 1 ohm	2
ST 19	ST,1,2, 16	DVM	01	28	-	Less than 1 ohm	2
ST 20	ST,4,16	DVM	57	01	-	405 to 495 ohms	2
ST 20	ST,4,16	DVM	58	01	-	405 to 495 ohms	2
ST 20	ST,4,16	DVM	55	01	MATRIX SELECTOR 1 : 12	90 to 110 ohms	2
ST 20	ST,4,16	DVM	56	01	MATRIX SELECTOR 2 : 12	90 to 110 ohms	2
ST 21	ST,1,4, 16	DVM	57	01	-	4.4 to 5.4K ohms	2
ST 21	ST,1,4, 16	DVM	58	01	-	4.4 to 5.4K ohms	2
ST 21	ST,1,4, 16	DVM	55	01	MATRIX SELECTOR 1 : 12	990 to 1210 ohms	2
ST 21	ST,1,4, 16	DVM	56	01	MATRIX SELECTOR 2 : 12	990 to 1210 ohms	2

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 22	ST,2,4, 16	DVM	57	01	-	22.4 to 27.4 ohms	2
ST 22	ST,2,4, 16	DVM	58	01	-	22.4 to 27.4 ohms	2
ST 22	ST,2,4, 16	DVM	55	01	MATRIX SELECTOR 1 : 12	10.0 to 12.2K ohms	2
ST 22	ST,2,4, 16	DVM	56	01	MATRIX SELECTOR 2 : 12	10.0 to 12.2K ohms	2
ST 23	ST,B,1, 2,4,16	DVM	01	-	-	Variable, 0 to -36 Vdc	1,5,8, 11,15
ST 23	ST,B,1, 2,4,16	DVM	02	-	-	+21.0 to +27.0 Vdc	1,8
ST 23	ST,B,1, 2,4,16	DVM	02	-	TEST/1/2 : TEST/2	-0.4 to +0.4 Vdc	1,8
ST 23	ST,B,1, 2,4,16	DVM	04	-	RESET/OFF/ON : RESET/ON	+21.0 to +27.0 Vdc	1,8
ST 23	ST,B,1, 2,4,16	DVM	04	-	-	-0.4 to +0.4 Vdc	1,8
ST 24	ST,8,16	DVM	01-58	-	-	1 megohm min.	1
ST 25	ST,1,8, 16	DVM	01-58	-	MATRIX SELECTOR 2 : 12 RESET/OFF/ON : RESET/ON	1 megohm min.	1
ST 26	ST,2,8, 16	DVM	01-08	-	-	1 megohm min.	1
ST 27	ST,1,2, 8,16	DVM	01-56	-	-	+4.5 to +5.5 Vdc	1,8
ST 28	ST,4,8, 16	DVM	01-56	-	-	+4.5 to +5.5 Vdc	1,8
ST 29	ST,1,4, 8,16	DVM	01-56	-	-	+4.5 to +5.5 Vdc	1,8

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 30	ST,2,4,8,16	DVM	01-56	-	-	+4.5 to +5.5 Vdc	1,8
ST 31	ST,1,2,4,8,16	O-Scope	01	01	-	T1 pulse (fig. 1)	8,13
ST 31	ST,1,2,4,8,16	O-Scope	02	01	-	$\overline{T1}$ pulse (fig. 1)	8,13
ST 31	ST,1,2,4,8,16	O-Scope	03	01	-	T2 pulse (fig. 1)	8,13
ST 31	ST,1,2,4,8,16	O-Scope	04	01	-	$\overline{T2}$ pulse (fig. 1)	8,13
ST 32	ST,32	O-Scope	01	01	-	T1 pulse (fig. 1)	8,13
ST 32	ST,32	O-Scope	02	01	-	T3 pulse (fig. 1)	8,13
ST 32	ST,32	O-Scope	03	01	-	$\overline{T3}$ pulse (fig. 1)	8,13
ST 32	ST,32	O-Scope	04	01	-	$\overline{T9}$ pulse (fig. 1)	8,13
ST 33	ST,1,32	O-Scope	19	19	-	$\overline{T1/4KC}$ pulse (fig. 1)	4,8,13
ST 33	ST,1,32	O-Scope	01	01	-	T1 pulse (fig. 1)	8,13
ST 33	ST,1,32	O-Scope	02	01	-	$\overline{T10}$ pulse (fig. 1)	8,13
ST 33	ST,1,32	O-Scope	03	01	-	$\overline{PRF}$ pulse (fig. 1)	8,13
ST 34	ST,2,32	DVM	05	24	OVERLOAD/OFF/ON : OVERLOAD/ON	Variable, 0 to +36 Vdc	2,5,9,10,15
ST 34	ST,2,32	DVM	05	24	-	+5.0 to +7.0 Vdc	2,9
ST 34	ST,2,32	DVM	06	25	-	+5.0 to +7.0 Vdc	2,9
ST 34	ST,2,32	DVM	07	26	-	+5.0 to +7.0 Vdc	2,9
ST 34	ST,2,32	DVM	12	22	-	-20.0 to -26.0 Vdc	2,9
ST 34	ST,2,32	DVM	13	23	-	-20.0 to -26.0 Vdc	2,9
ST 34	ST,2,32	DVM	16	34	-	+20.0 to +26.0 Vdc	2,9
ST 34	ST,2,32	DVM	17	35	-	+20.0 to +26.0 Vdc	2,9
ST 34	ST,2,32	DVM	41	51	-	-28.0 to -38.0 Vdc	2,9
ST 34	ST,2,32	DVM	42	52	-	-28.0 to -38.0 Vdc	2,9
ST 34	ST,2,32	DVM	45	53	-	+28.0 to +38.0 Vdc	2,9
ST 34	ST,2,32	DVM	46	54	-	+28.0 to +38.0 Vdc	2,9

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 34	ST,2,32	DVM	10	11	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	14	15	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	20	21	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	36	37	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	43	44	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	47	48	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	49	50	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	55	56	-	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	16	14	-	Resistance ratio 10:1 or greater	2,6,7
ST 34	ST,2,32	DVM	14	36	-	Resistance ratio 10:1 or greater	2,6,7
			36	14			
ST 34	ST,2,32	DVM	16	36	OVERLOAD/OFF/ON : OVERLOAD/ON	Less than 1 ohm	2,6
ST 34	ST,3,32	DVM	12	10	-	Resistance ratio 10:1 or greater	2,6,7
			10	12			
ST 34	ST,2,32	DVM	12	20	OVERLOAD/OFF/ON : OVERLOAD/ON	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	10	20	-	Resistance ratio 10:1 or greater	2,6,7
			20	10			
ST 34	ST,2,32	DVM	45	47	-	Resistance ratio 10:1 or greater	2,6,7
			47	45			
ST 34	ST,2,32	DVM	47	55	-	Resistance ratio 10:1 or greater	2,6,7
			55	47			
ST 34	ST,2,32	DVM	45	55	OVERLOAD/OFF/ON : OVERLOAD/ON	Less than 1 ohm	2,6
ST 34	ST,2,32	DVM	41	43	-	Resistance ratio 10:1 or greater	2,6,7
			43	41			
ST 34	ST,2,32	DVM	43	49	-	Resistance ratio 10:1 or greater	2,6,7
			49	43			

* See notes at end of Table.

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TABLE I. Self test card output.-continued

Test Card	Test Ident	Test Equip.	Test Points A	Test Points B	Exception to Switches Selected	Output	Notes
ST 34	ST,2,32	DVM	41	49	OVERLOAD/OFF/ON : OVERLOAD/ON	Less than 1 ohm	2,6
ST 35	ST,1,2, 32	DVM	30	24	-	2.75 to 4.75 ohms	2,6
ST 35	ST,1,2, 32	DVM	31	24	-	2.75 to 4.75 ohms	2,6
ST 35	ST,1,2, 32	DVM	32	24	-	2.75 to 4.75 ohms	2,6
ST 35	ST,1,2, 32	DVM	20	22	-	27.2 to 33.2 ohms	2,6
ST 35	ST,1,2, 32	DVM	21	22	-	27.2 to 33.2 ohms	2,6
ST 35	ST,1,2, 32	DVM	36	22	-	27.2 to 33.2 ohms	2,6
ST 35	ST,1,2, 32	DVM	37	22	-	27.2 to 33.2 ohms	2,6
ST 35	ST,1,2, 32	DVM	49	51	-	45 to 55 ohms	2,6
ST 35	ST,1,2, 32	DVM	50	51	-	45 to 55 ohms	2,6
ST 35	ST,1,2, 32	DVM	55	53	-	45 to 55 ohms	2,6
ST 35	ST,1,2, 32	DVM	56	53	-	45 to 55 ohms	2,6

* See notes at end of Table.

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TABLE I. Self test card output (Notes).

NOTES:

1. Connect plus lead of multimeter to TEST POINT A and negative lead to GRD 1.
2. When TEST POINT A and TEST POINT B selections are indicated for multimeter checks connect plus lead to TEST POINT A and negative lead to TEST POINT B.
3. Connect plus lead of multimeter to TEST POINT B and negative lead to GRD 1.
4. Repetition rate of this pulse is 4 kHz.
5. Connect Power Supply, item 11 to DC IN connector via cable W85, item 24. Connector terminal with red stripe should be connected to + (red terminal) of Power Supply, item 11. Voltage variable via Power Supply, item 11.
6. Open +6.6 V, +15 V, -15V, +25 V, and -25 V circuit breakers.
7. Resistance measurements taken across a tranistor. Check that resistance measurement is at least 10 times greater in one direction than in the other. Do not set multimeter scale below 1k ohm.
8. Connect a 3k ohm resistor CP32, item 21, between TEST POINT A and GRD 1 for these tests.
9. Connect a 3k ohm resistor CP37, item 22, between TEST POINT A and TEST POINT B for these tests.
10. Amplitude adjustable by Power Supply, item 11. Limiting occurs at +37 to +41 Vdc.
11. Amplitude adjustable by Power Supply, item 11. Limiting occurs at -37 to -41 Vdc.
12. Set a TRIGGERING SOURCE switch on oscilloscope to INT.
13. Use test set-up as shown in Figure 2.
14. Function Generator set-up: FREQUENCY X1, frequency dial as noted, SWEEP WIDTH maximum CCW, SWEEP RATE in MANUAL position (press in knob), MAIN OUTPUT in REAR position, OFFSET switch adjusted to zero. POWER to ON.
15. Remove ground strap from Power Supply, item 11.
16. Connect a 3k ohm resistor CP32, item 21, between TEST POINT B and GRD 1 for these tests.

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4.4.3 Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and re-submittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.5 Special sampling. One printed circuit card tester out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table II and shall then meet the requirements and tests in Table III.

TABLE II. Classification of defects

100% Inspection.

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Electrical	3.4	4.7.2
102.	POWER/ON/OFF switch S1	3.5.1	4.7.3.1
103.	Circuit breakers	3.5.2	4.7.3.2
104.	LEVEL/AC/DC switch S2	3.5.3	4.7.3.3
105.	OVERLOAD/OFF/ON switch S7	3.5.4	4.7.3.4
106.	PROFILE/OFF/ON switch S3	3.5.5	4.7.3.5
107.	NORMAL/HIGH/LOW switch S4	3.5.6	4.7.3.6
108.	TEST/1/2 switch S5	3.5.7	4.7.3.7
109.	RESET/OFF/ON switch S6	3.5.8	4.7.3.8
110.	Power indicators	3.5.9	4.7.3.9
111.	Digiswitch lamps	3.5.10	4.7.3.10
112.	Test points A and B, and TEST IDENTIFICATION lamp outputs	3.5.11	4.7.3.11

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and a resubmitted sample has been approved. (See 4.4.3).

TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Fabrication	3.1	Applicable drawings - visual
302. General specification	3.2	MIL-F-13926/ MIL-C-46357 - visual
303. Vibration	3.3.1	4.7.1.1
304. Shock	3.3.2	4.7.1.2
305. Altitude - operating	3.3.3.1	4.7.1.3
306. Altitude - non-operating	3.3.3.2	4.7.1.3
307. Temperature	3.3.4	4.7.1.4
308. Humidity - non-operating	3.3.5	4.7.1.5

4.6 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the referenced specifications and standards cited in 4.4.2. Inspection equipment shall be as listed in 4.6.1. The contractor shall submit inspection equipment designs for approval in accordance with the terms of the contract. See 6.3 of this specification.

4.6.1 Inspection equipment. The following inspection equipment or equivalent shall be used in the performance of the applicable test as specified in 4.7.

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TABLE IV. Inspection equipment list.

Item Number	Description
1. Ohmmeter	To measure resistance, zero ohms to greater than 1 megohm
2. Voltmeter	DC, 0 to 50 volts, accuracy of 1%
3. Voltmeter	AC, 0 to 150 volts, accuracy of 1%
4. Power Supply	DC, 24 volts ± 1%, 6A min.
5. Power Supply	DC, 33 volts ± 1%, 2A min.
6. Power Supply	DC, 33 volts ± 1%, 2A min.
7. Power Supply	DC, 23 volts ± 1%, 2A min.
8. Power Supply	DC, 23 volts ± 1%, 2A min.
9. Power Supply	DC, 5.9 volts ± 1%, 6A min.
10. Power Supply	AC, 115 volts ± 10% 400 Hz ± 5 Hz, 2A min.
11. Power Supply	11727789
12. Oscilloscope	Vertical sensitivity = 0.2 to 2.0V/div. Horizontal deflection = 0.1 usec/div. Vertical pass band - DC to 10 MHz.
13. Resistor	17 ohms ± 10%, not less than 68 W
14. Resistor	12 ohms ± 10%, not less than 48 W
15. Resistor	1 ohm ± 10%, not less than 36 W
16. Resistor	4 ohms ± 10%, not less than 144 W
17. Resistor	63 ohms ± 10%, not less than 252 W

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TABLE IV. Inspection equipment list, continued

Item Number		Description
18.	Adapter CP4	11726394
19.	Adapter CP6	11726396
20.	Adapter CP13	10556106
21.	Adapter CP32	11740292
22.	Adapter CP37	10554918-2
23.	Test Cards	11726471
24.	Cable Assy, W85	11727743

4.7 Methods of inspection.

4.7.1 Environmental. Environmental test shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 except as specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Procedure I, Category 1, of MIL-STD-810 except as modified by paragraph 3.3.1 herein. After vibration, the printed circuit card tester shall be tested as given in the sub-paragraph of 4.7.2 and 4.7.3 for compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I of Method 516 of MIL-STD-810 except that shock pulses shall be as specified in 3.3.2. After the shock impulses, the printed circuit card tester shall be tested as specified in 4.7.2 and 4.7.3 for compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedure I and II, Method 501, of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedure I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7 the operational checkout shall be performed at the end of the tenth cycle only at standard ambient conditions.

4.7.2 Electrical. Continuity shall be inspected by ohmmeter item 1 and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 POWER/ON/OFF switch S1.

4.7.3.1.1 POWER/OFF legend. Apply 24.0 ± 1.2 Vdc using item 4 between J1-U(+) and J1-T(Ret). Set S1 to normally closed (OUT) position. Verify that POWER/OFF legend is illuminated.

4.7.3.1.2 POWER/ON legend. Under conditions as stated in 4.7.3.1.1, operate S1 to the normally open (IN) position and verify that POWER/ON legend is illuminated.

4.7.3.2 Circuit breakers. Apply voltages to J1 as follows:

a. 33.0 ± 0.3 Vdc between J1-A(-) and J1-B(Ret) using item 5.

b. 33.0 ± 0.3 Vdc between J1-C(+) and J1-D(Ret) using item 6.

c. 23.0 ± 0.2 Vdc between J1-E(-) and J1-F(Ret) using item 7.

d. 23.0 ± 0.2 Vdc between J1-G(+) and J1-F(Ret) using item 8.

e. 5.90 ± 0.06 Vdc between J1-P(+) and J1-R(Ret) using item 9.

f. 115.0 ± 5.0 Vac, 400 ± 5 Hz between J1-V and J1-W using item 10.

g. 24.0 ± 0.2 Vdc between J1-S(+) and J1-T(Ret) using item 4.

Use voltmeter, item 2, to verify that circuit breakers CB1 through CB7 are closed when reset buttons are depressed so white bands are concealed, and open when reset buttons are pulled out to expose white bands. Use voltmeter, item 3, to perform same test on circuit breaker CB8. With circuit breakers CB1 through CB8 closed, use resistors, items 13 through 17, to check overload protection of circuit breakers as follows. Connect resistor indicated below from load side of circuit breaker to ground, and verify that breaker trips within 20 seconds.

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Circuit Breaker	Resistor Load
-25V (CB1)	Item 13
+25V (CB2)	Item 13
-15V (CB3)	Item 14
+15V (CB4)	Item 14
+6.6V (CB5)	Item 15
+24V EXT (CB6)	Item 14
+24V INT (CB7)	Item 16
LINE (CB8)	Item 17

NOTE: These tests may be performed prior to the installation of the circuit breaker into the next higher assembly when verification of the test is provided.

4.7.3.3 LEVEL/AC/DC switch S2.

4.7.3.3.1 LEVEL/DC legend. Apply 24.0 ± 1.2 Vdc using item 4 between J1-S(+) and J1-T(Ret). Set up conditions as stated in 3.5.1.2. Close +24V INT circuit breaker (CB7) and operate S2 to normally closed (OUT) position. Verify that LEVEL/DC legend is illuminated.

4.7.3.3.2 LEVEL/AC legend. Under conditions as stated in 4.7.3.3.1 operate S2 to the normally open (IN) position and verify that LEVEL/AC legend is illuminated.

4.7.3.4 OVERLAOD/OFF/ON switch S7.

4.7.3.4.1 OVERLOAD/OFF legend. Set up conditions as stated in 3.5.3.1a. through 3.5.3.1c. With S7 in the normally closed (OUT) position, verify that OVERLOAD/OFF legend is illuminated.

4.7.3.4.2 OVERLOAD/ON legend. Set up conditions as stated in 4.7.3.4.1, depress and hold S7 to normally open (IN) position and verify that OVERLOAD/ON legend is illuminated.

4.7.3.5 PROFILE/OFF/ON switch S3.

4.7.3.5.1 PROFILE/OFF legend. Set up conditions as stated in 3.5.3.1a. through 3.5.3.1c. Set S3 to normally closed (OUT) position and verify that PROFILE/OFF legend is illuminated.

4.7.3.5.2 PROFILE/ON legend. Set up conditions as stated in 4.7.3.5.1, operate S3 to normally open (IN) position and verify that PROFILE/ON legend is illuminated.

4.7.3.6 NORMAL/HIGH/LOW switch S4.

4.7.3.6.1 NORMAL legend. Set up conditions as stated in 3.5.3.1a. through 3.5.3.1c. Set PROFILE/OFF/ON switch S3 to illuminate PROFILE/OFF legend. Verify that NORMAL legend of S4 is illuminated.

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4.7.3.6.2 HIGH legend. Set up conditions as stated in 4.7.3.6.1. Operate PROFILE/OFF/ON switch S3 to illuminate PROFILE/ON legend. Set S4 to normally closed (OUT) position and verify that HIGH legend of S4 is illuminated.

4.7.3.6.3 LOW legend. Set up conditions as stated in 4.7.3.6.2. Operate S4 to normally open (IN) position and verify that LOW legend of S4 is illuminated.

4.7.3.7 TEST/1/2 switch S5.

4.7.3.7.1 TEST/1 legend. Set up conditions as stated in 3.5.3.1a. through 3.5.3.1c. Set S5 to the normally closed (OUT) position and verify that TEST/1 legend of S5 is illuminated.

4.7.3.7.2 TEST/2 legend. Set up conditions as stated in 4.7.3.7.1. Operate S5 to the normally open (IN) position and verify that TEST/2 legend of S5 is illuminated.

4.7.3.8 RESET/OFF/ON switch S6.

4.7.3.8.1 RESET/OFF legend. Set up conditions as stated in 3.5.3.1a. through 3.5.3.1c. Set S6 to normally closed (OUT) position. Insert Adapter CP6, item 19, into test sockets A and B. Insert Self Test Card ST1 into CARD READER ASSEMBLY and operate handle. Verify that RESET/OFF legend of S6 is illuminated.

4.7.3.8.2 RESET/ON legend. Set up conditions as stated in 4.7.3.8.1. Operate S6 to normally open (IN) position and verify that RESET/ON legend of S6 is illuminated.

4.7.3.9 POWER indicators. Set up conditions as stated in 3.5.1.2. Apply voltages as in 4.7.3.2a. through 4.7.3.2g. with CB1 through CB8 closed (no white band showing). Insert Adapter CP6, item 19, into test sockets A and B and insert Self Test Card ST1 into the CARD READER ASSEMBLY. Verify that power indicators +6.6V, -15V, +15V, +24V EXT, +24V INT, -25V, +25V and LINE are illuminated.

4.7.3.10 Digiswitch Lamps. Set up conditions stated in 4.7.3.9. Verify that two thumbwheel switch lamps are illuminated in each of TEST POINT A and TEST POINT B groups in conformance with 3.5.10.

4.7.3.11 Test points A and B, and TEST IDENTIFICATION Lamp Outputs. Set up conditions stated in 3.5.11 and 4.7.3.2a. through 4.7.3.2g. Insert Self Test Card ST1 into Card Reader Assembly. Using items 1, 2, 3 and 12, verify that outputs exist as stated in Table I and 3.5.11. Remove Self Test Card ST1 and verify all outputs in Table I and 3.5.11 by inserting Self Test Cards ST2 through ST33 in turn. Remove Adapter CP6, item 19, and insert Adapter CP4, item 18, into test sockets B and C. Verify outputs for Self Test Cards ST34 and ST35.

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5. PACKAGING

5.1 General. Preservation, packing, and marking shall be in accordance with Packaging Data Sheet 11726306.

6. NOTES

6.1 Intended use. The printed circuit card tester is a part of Radar Test Set AN/TPM-22A. It is used to check the performance of and assist in isolating malfunctions in printed circuit cards used in Radar Test Set AN/TPM-22A and in Radar Test Set AN/VPS-2.

6.2 Ordering data. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this and the item detail specifications.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Quality Conformance Inspection, if other than specified in Section 4 of item detail specification.

e. First Article Sample requirements, if other than specified in Section 4 of item detail specification.

f. Packaging requirements, if other than specified in Section 5 of item detail specification.

g. Certificate of conformance for each lot or shipment of product.

6.2.2 Contract data requirements.

a. Contract data requirements for inspection equipment designs conforming to Data Item Description DI-R-1714, tailored, with appropriate addendum to each. See 6.3.

b. Data Cards - When required by incoroporation of DD Form 1423 into the contract, data cards shall be prepared for each lot in accordance with Data Item Description DI-L-1410.

6.3 Inspection equipment designs. Inspection equipment designs are of two types - Government designs and contractor designs. Government designs are designated by drawing numbers in the appropriate paragraph in Section 4.

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Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government design is provided; however, unless otherwise specified, the contractor may submit alternate or modified contractor designs in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 Government designs. Government designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in DOD-STD-100.

c. An envelope drawing, as defined in DOD-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for All inspection equipment for which Government designs are not specified and include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with Government designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per DOD-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The item detail specification number, paragraph number and defect number from Section 4 shall be referenced on each contractor design together with the component or assembly drawing number, revision letter and date to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements Lists (DD Form 1423). Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAF-I (D), Dover, NJ 07801.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as US Army Armament Research and Development Command, (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Changes to previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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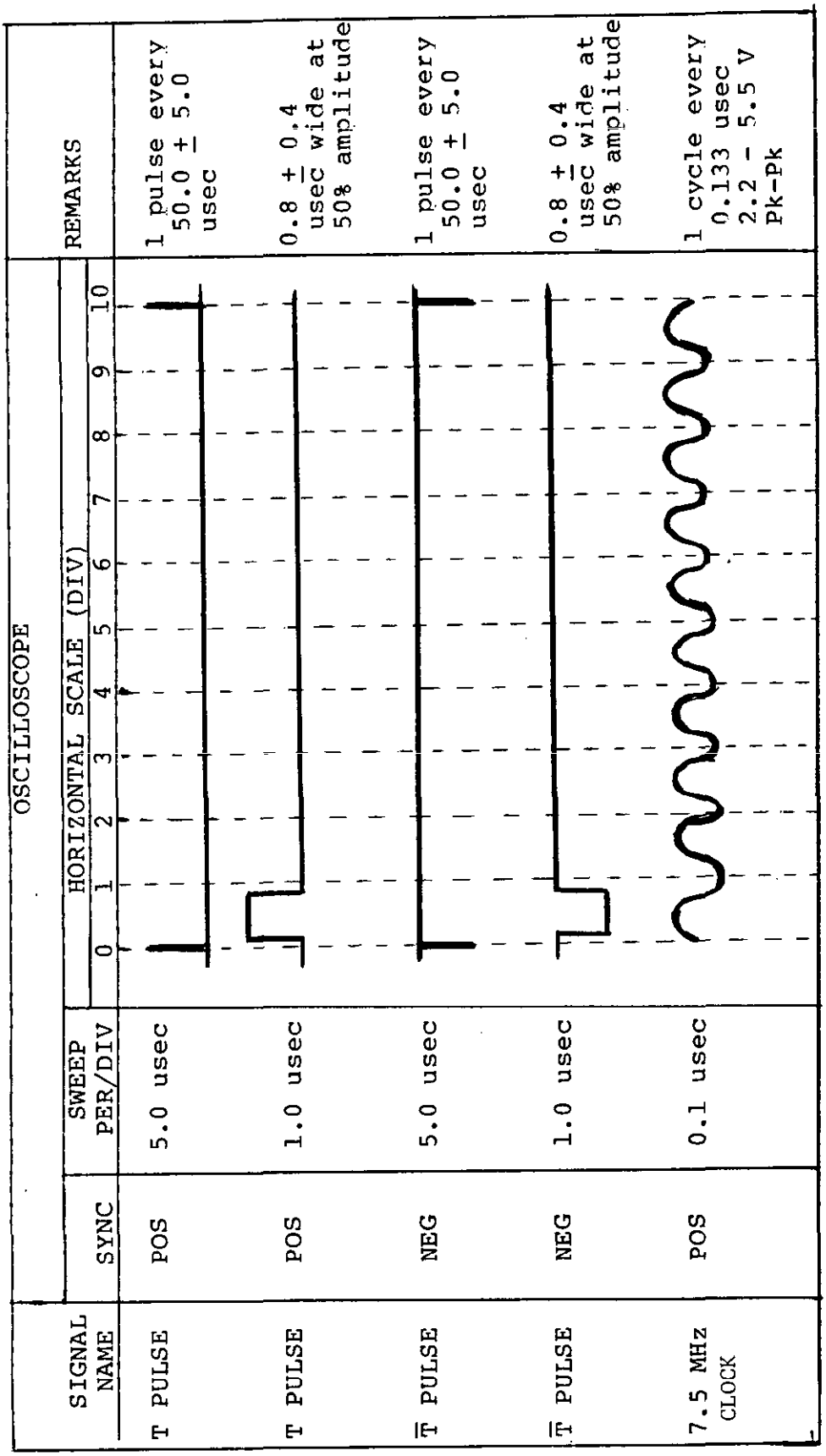


FIGURE 1. Waveforms.

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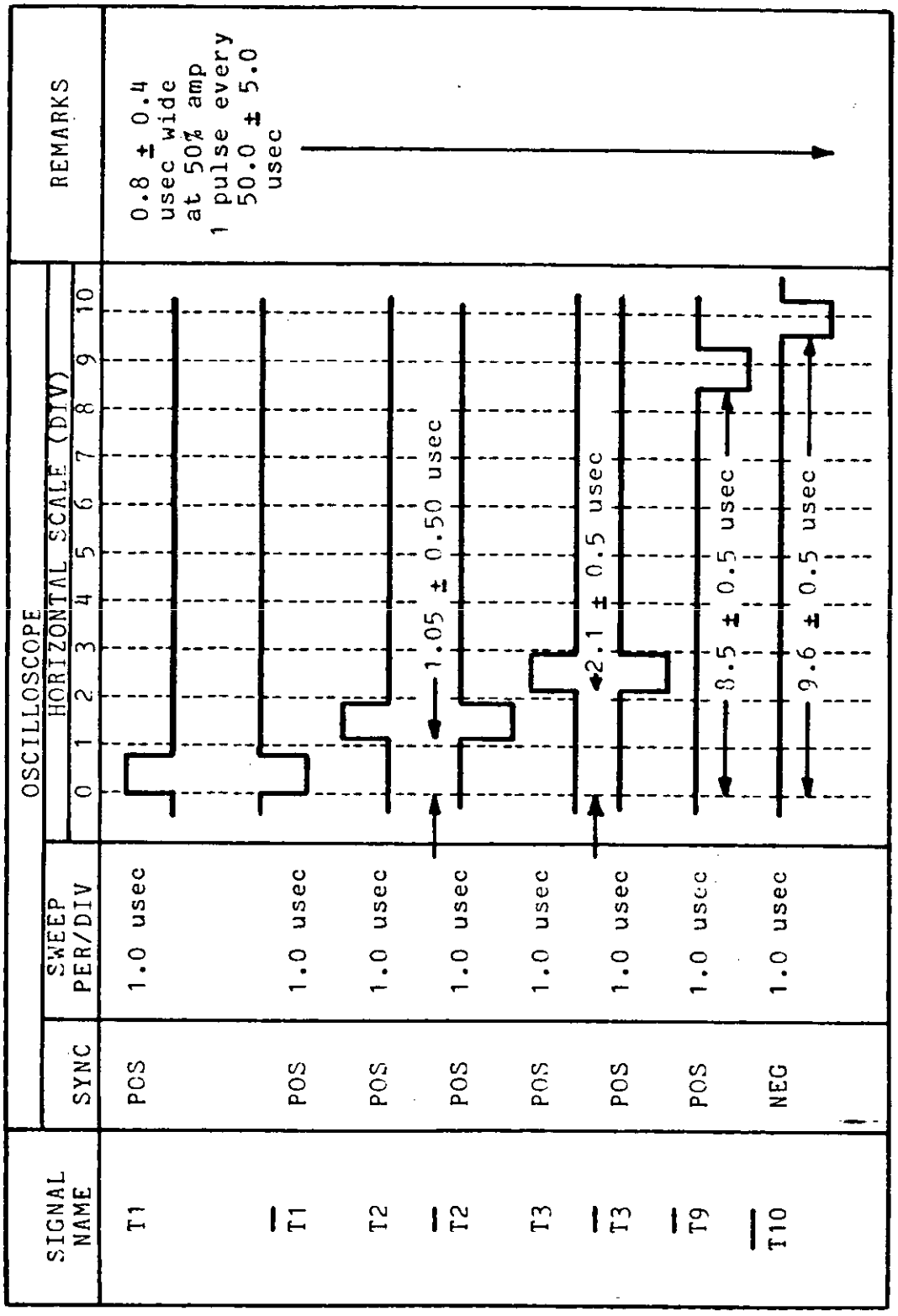


Figure 1. Waveforms - continued

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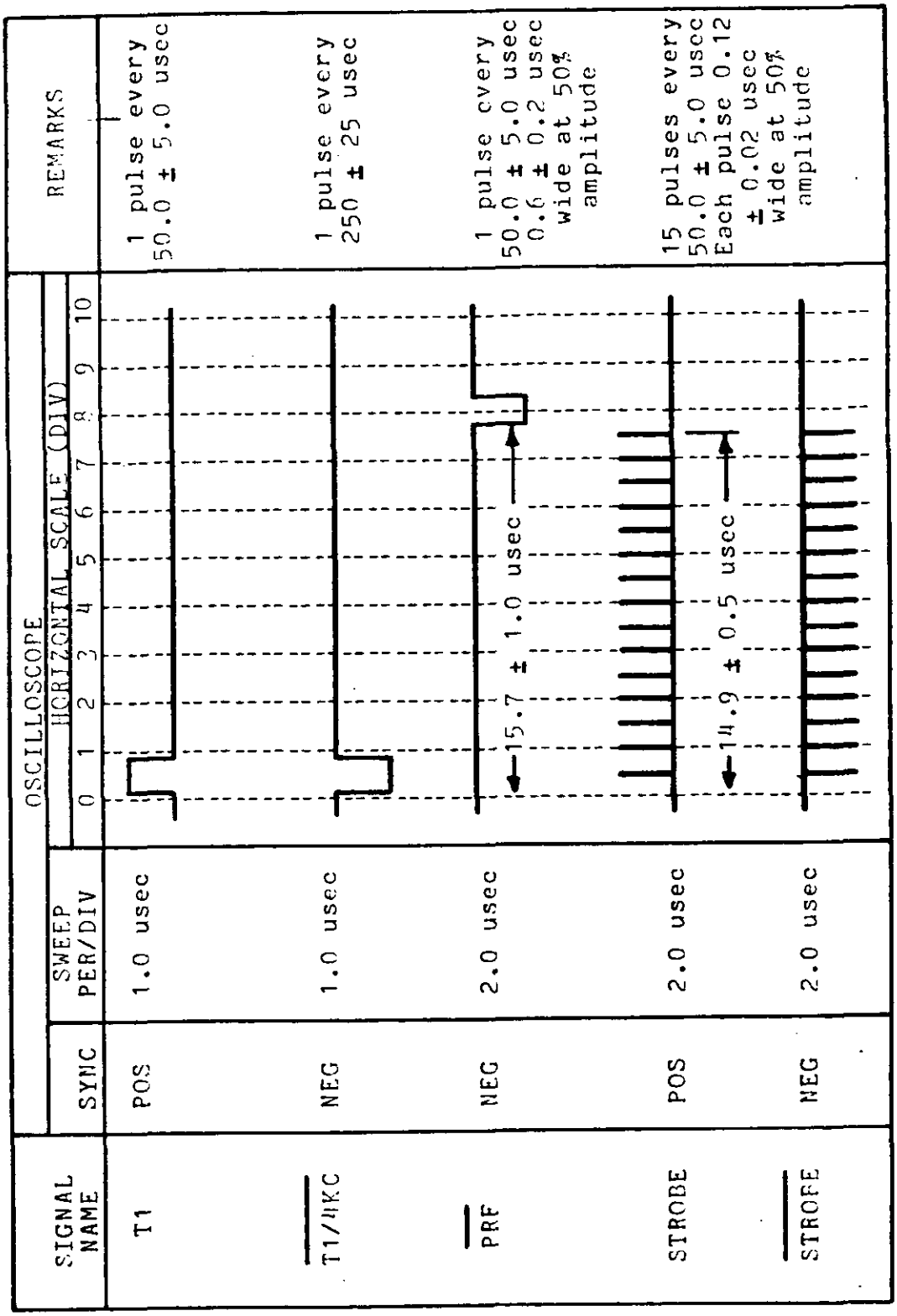


Figure 1. Waveforms - continued

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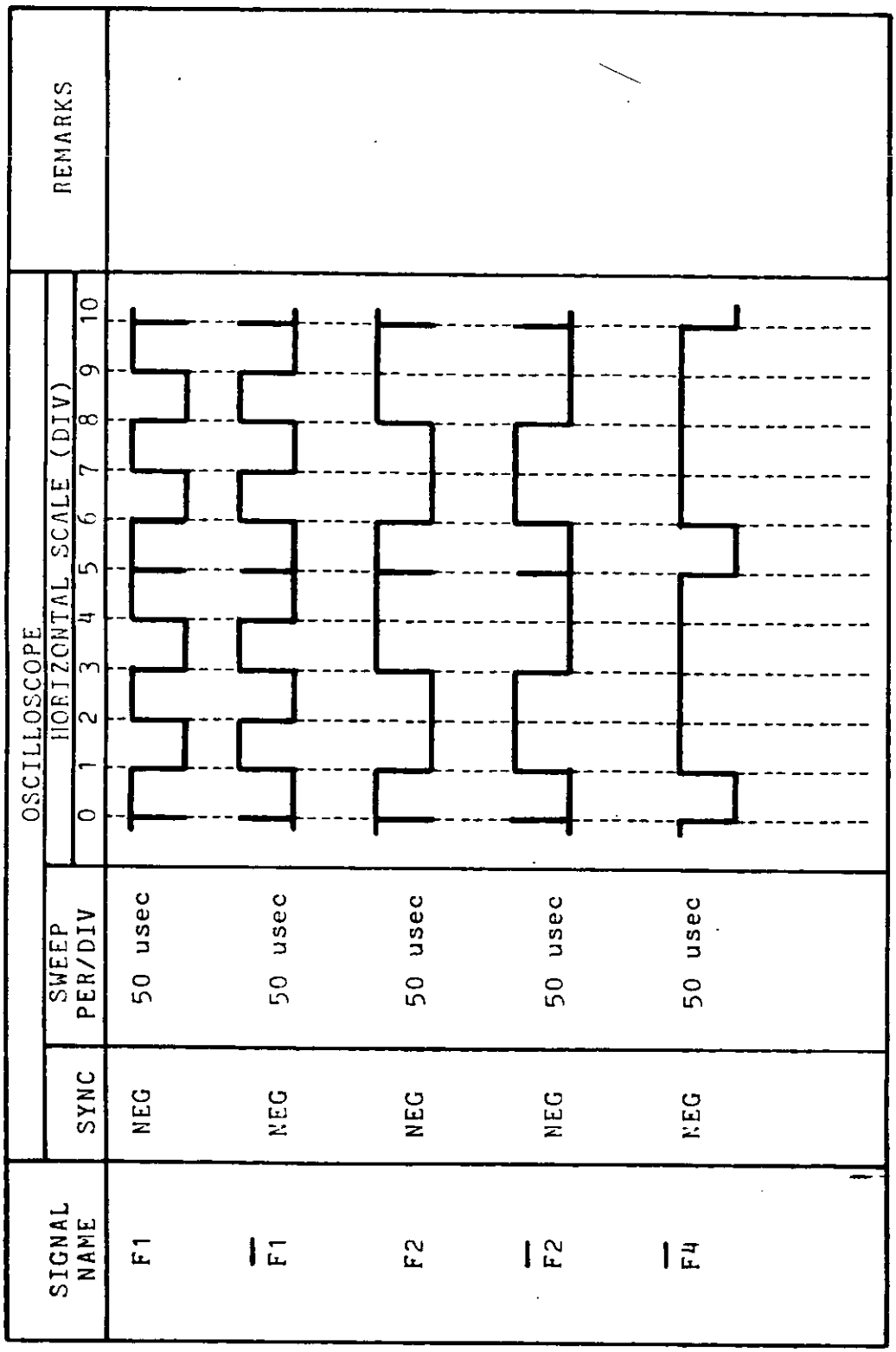


Figure 1. Waveforms - continued

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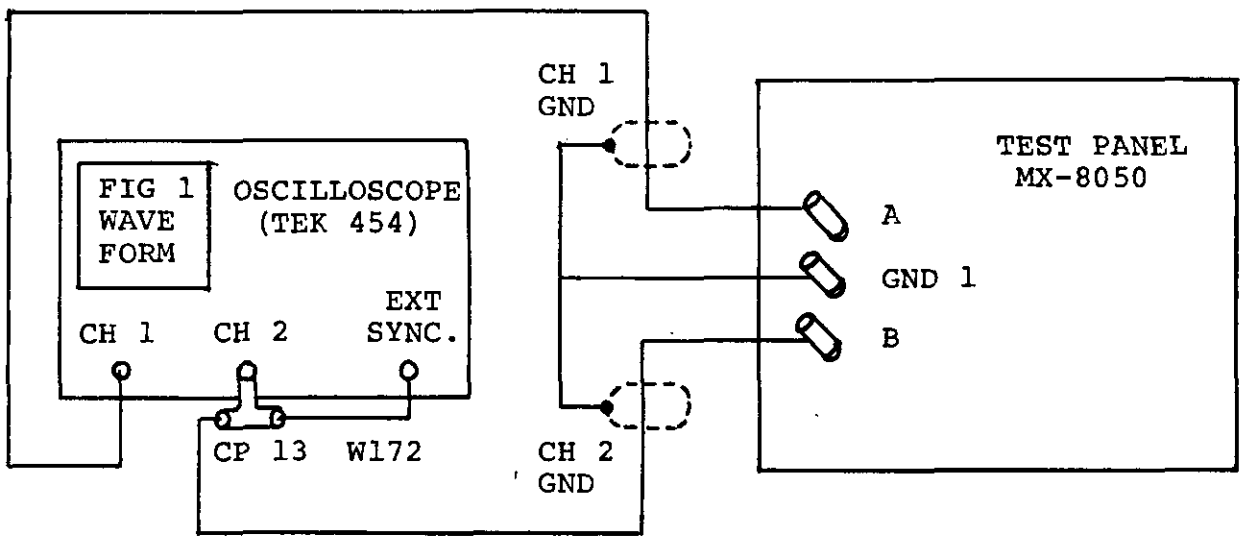


Figure 2. Test set-up.

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-50339A (AR)	Test Set, Subassembly MX-8050/TPM-22A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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